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ACTIVE CABLE WITH REMOTE END CONTROL ACCESS

Abstract

Cable designs and methods are provided herein to enable remote end access to active cable controllers for monitoring and upgrade operations. These designs and methods enable a firmware update method that includes: downloading firmware from a host to a local controller in a local end connector of a cable, the local controller storing the firmware in a local memory; transferring a copy of the firmware via an internal channel from the local controller to a remote controller in a remote end connector of the cable, the remote controller storing the firmware in a remote memory in the remote end connector; and configuring the local controller and the remote controller to each boot from the firmware when the cable is restarted.

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Background/Summary

BACKGROUND

[0001] Data centers for cloud computing tend to be large, complex installations having hundreds of thousands servers interconnected by routers and a correspondingly large number of network cables, each cable being asked to support data transfer rates near the limits of what is currently achievable. To enhance their performance, so-called “active” cables typically rely on embedded electronics that perform signal processing on the received signals (and often on the signals to be transmitted as well). Some cable manufacturers make the embedded electronics configurable, enabling the cables to be upgraded as new processing techniques are tested and proven to yield better performance. However, given the number of cables in a typical data center, the upgrade process can be quite challenging.

SUMMARY

[0002] Accordingly, there are disclosed herein novel cable designs and methods enabling remote end access to active cable controllers for monitoring and upgrade operations. These designs and methods enable a firmware update method that includes: downloading firmware from a host to a local controller in a local end connector of a cable, the local controller storing the firmware in a local memory; transferring a copy of the firmware via an internal channel from the local controller to a remote controller in a remote end connector of the cable, the remote controller storing the firmware in a remote memory in the remote end connector; and configuring the local controller and the remote controller to each boot from the firmware when the cable is restarted.

[0003] In an illustrative network having multiple network nodes including one or more switches, and active network cables that interconnect the multiple network nodes, each of the switches can be configured to use the foregoing method to perform a firmware update in each connected one of the active network cables. Each switch may provide a monitoring and maintenance service that can perform the firmware update without burdening the non-switch network nodes.

[0004] Each of the foregoing may be implemented individually or in combination, and may be implemented with one or more of the following features in any suitable combination: 1. no non-switch network nodes are configured to perform the firmware update. 2. as part of performing the firmware update each of the switches is further configured to initiate copying of the firmware via the internal channel from the local controller to a second remote controller in a second remote end connector of the cable, the second remote controller storing the firmware in a second remote memory. 3. as part of performing the firmware update each of the switches is further configured to access at least one of the local controller and the remote controller to determine whether an installed firmware version is current. 4. as part of performing the firmware update each of the switches is further configured to initiate a restart of that cable.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 shows an illustrative network.

[0006] FIG. 2 is an isometric view of an illustrative two-ended active cable.

[0007] FIG. 3 is an isometric view of an illustrative three-ended active cable.

[0008] FIG. 4 is a block diagram of an illustrative two-ended active cable.

[0009] FIG. 5 is a block diagram of an illustrative three-ended active cable.

[0010] FIG. 6 is a block diagram of a transmit chain in an illustrative multi-lane transceiver.

[0011] FIG. 7 is a block diagram of a receive chain in an illustrative multi-lane transceiver.

[0012] FIG. 8 is a flow diagram of an illustrative method for remotely upgrading cable firmware.

DETAILED DESCRIPTION

[0013] While specific embodiments are given in the drawings and the following description, keep in mind that they do not limit the disclosure. On the contrary, they provide the foundation for one of ordinary skill to discern the alternative forms, equivalents, and modifications that are encompassed in the scope of the appended claims.

[0014] FIG. 1 shows an illustrative network **100**, which includes three network nodes **102**, **103**, **104**, and three switches **106**, **107**, **108**. The network nodes **102-104** each represent a device capable of sending and receiving communications via a network, such as a server, a storage device, a workstation, a phone, a printer, a scanner, a network hub, a network bridge, a switch, a router, or any device having a network port. The Ethernet Standard (IEEE Std 802.3-2015 or one of its updates) is used as an example herein, but any wired, optical, or cabled network standard would also be suitable.

[0015] Switches **106-108** are each a device having multiple network ports and an internal mechanism for directing messages received on one of the network ports to another of the network ports. As used hereinafter, the term “switch” includes not just traditional network switches, but also routers and network bridges. Frequently one or more of the switch ports **109** connect to other switches to enable communication between the nodes **102-104** and a wider-area network such as the Internet.

[0016] The illustrative network includes two-ended network cables **110** connecting two network ports of switch **107** to corresponding network ports of switch **108**, and cables **112** connecting two ports of switch **106** to corresponding ports of switch **108**. A three-ended cable **114** connects a network port of node **102** to one port of switch **106** and one port of switch **107**. Conversely, a three-ended cable **116** connects one switch port **106** to a port of node **103** and a port of node **104**. Three-ended cables **114**, **116** may be conventional breakout cables or enhanced cables offering two of the three ends for redundant connections to third, shared end, such that each cable connector can support the full data stream bandwidth, enabling one of the ports to be bypassed when a fault is detected.

[0017] FIG. 2 shows an illustrative two-ended cable having a first end **201** connected to a second end **202** via a cord **206** having electrical conductors and/or optical fibers for digital communications signals to convey data streams in each direction pursuant to network communications standard such as the Institute of Electrical and Electronics Engineers (IEEE) Standard for Ethernet, IEEE Std 802.3-2015 or any of its periodic updates, which will be familiar to those of ordinary skill in the art to which this application pertains. To enable communication at higher bandwidths and/or over extended cable lengths, either or both connectors **201**, **202** may include a powered transceiver that preferably performs clock and data recovery (CDR) and remodulation of data streams flowing in each direction. These powered transceivers are also known as data recovery and remodulation (DRR) devices. Notably, these transceivers may perform CDR and remodulation not only of the outbound data streams as they exit the cable, but also of the inbound data streams as they enter the cable. In certain alternative embodiments, the powered transceivers provide amplification, equalization, and/or conversion between electrical and optical signals, without performing CDR.

[0018] In some contemplated implementations, the cable connectors **201**, **202** are quad small form-factor pluggable (QSFP) transceiver modules, and more specifically, QSFP28 transceiver modules, which exchange CAUI-4 data streams with the host. In other contemplated implementations, the cable connectors are dual small form-factor pluggable (DSFP) or small form-factor pluggable, double-density (SFP-DD) transceiver modules, which exchange 100GBASE-KR2 data streams with the host. (More properly, this SFP-DD transceiver is called SFP56-DD.) In still other

contemplated implementations, the cable connectors are different, such as with a QSFP28 to SFP56-DD cable, a QSFP28 to DSFP cable, or a SFP56-DD to DSFP cable.

[0019] FIG. 3 is an isometric view of an illustrative three-ended cable having a first, non-redundant connector **301** connected to a second and third redundant connectors **302**, **303** by electrical, or optionally by optical, conductors **306**. Either or both the non-redundant connector **301** and redundant connectors **302**, **303** may contain powered transceivers to improve performance. If implemented as a breakout cable, the data streams conveyed by the split ends **302**, **303** are merged at the unary end **301**. Though only two-ended and three-ended cables have been shown, we note that the principles discussed herein are applicable to N-ended cables where N is any integer greater than 2. (1:3 and 1:4 cables are specifically contemplated.)

[0020] FIG. 4 is a function-block diagram of an illustrative two-ended cable. Connector **201** is adapted to fit a standard-compliant Ethernet port in a first host device to receive an outbound data stream from the host device and to provide an inbound data stream to the host device via electrical contacts for one or more bidirectional data lanes (LN0, LN1). Similarly, connector **202** fits an Ethernet port of a second host device. Connector **201** includes a first DRR device **402** to perform CDR and remodulation of the data streams entering and exiting the cable at connector **201**, and connector **202** includes a second DRR device **404** to perform CDR and re-modulation of the data streams entering and exiting the cable at connector **202**. The DRR devices **402**, **404** may be integrated circuits mounted on a printed circuit board and connected to edge connector contacts via circuit board traces. Electrical conductors **406** (or alternatively, optical fibers) convey the data streams transiting between DRR devices **402**, **404**.

[0021] In at least some contemplated implementations, the printed circuit boards each also support a microcontroller unit (MCU) **403**, **405**. Each DRR device **402**, **404** is coupled to a respective MCU device **403**, **405**, which configures the operation of the DRR device via a first two-wire bus. At power-on, the MCU devices **403**, **405** load settings from a nonvolatile memory **414** into the DRR device's registers **412** to configure operation of the DRR's internal controller **410**. The host device can access the local MCU device **403**, **405** via a second two-wire bus that operates in accordance with the I2C bus protocol and/or the faster MDIO protocol. With this access to the local MCU device **403**, **405**, the host device can adjust the cable's local operating parameters and monitor the cable's performance.

[0022] FIG. 4 further shows electrical conductors **408** for management bus communications between the DRR devices **402**, **404**. In FIG. 4, the conductors support unidirectional serial communications between UART (universal asynchronous receiver-transmitter) modules **416**, but other management bus implementations would also be suitable. With the management bus connecting the MCUs, each host device can access not only the local MCU to set operating parameters and monitor performance, but also the MCU in the remote end connector(s). As discussed further below, this feature is expected to facilitate cable performance monitoring and upgrading operations in data centers.

[0023] FIG. 5 is a function-block diagram of an illustrative three-ended cable. Non-redundant connector **301** includes a DRR device **402** coupled to MCU **403**. Channel conductors **506** couple the DRR device **402** to electrical contacts in the remote end connectors **302**, **303**. Management bus conductors **508** also couple the MCU **403** to management bus contacts in the remote end connectors **302**, **303**. FIG. 5 shows the management bus as a shared I2C bus that enables each of the hosts to arbitrate for access to the MCU **403**. Alternatively, the MCU **403** could be coupled to each end connector by point-to-point management bus conductors to support, e.g., I2C or MDIO links.

[0024] Other contemplated cable implementations eliminate any need for dedicated management bus conductors along the length of the cable by, e.g., implementing a hidden channel within the transit data streams. To illustrate how this might be done, FIGS. 6 and 7 provide a more detailed block diagram of transmit and receive chains designed to couple with a circuit using one of the

physical coding sublayer (PCS) interface protocols specified in the Ethernet Standard. Pursuant to the standard, the PCS data stream is already encoded with a transmission code that provides DC balance and enables timing recovery. The PCS data stream lanes further include PCS alignment markers for synchronizing the lanes with each other.

[0025] Once the data stream lanes are aligned, an alignment marker removal module **602** removes the alignment markers from each lane, passing them to a downstream alignment marker insertion module **606**. A transcoding module **604** modifies the transmission code from a 64b/66b code to a 256b/257b code more appropriate for use with the Reed-Solomon encoder. By repeatedly transcoding four 66-bit blocks taken in parallel from the four incoming lanes into individual 257-bit blocks, the transcoding module may essentially convert a multiple lane data stream into a single lane data stream.

[0026] The previously-mentioned alignment marker insertion module **606** accepts the PCS alignment marker information from removal module **602** and the single-lane data stream from transcoding module **604**. The insertion module **606** combines the alignment marker information from the four lanes into a set of grouped alignment markers in a 257-bit block and, accounting for the operation of the transcoding module **604**, inserts the alignment block in a fashion that preserves its location relative to the other data in the data stream **607**. The alignment marker insertion block is designed to account for the operation of the encoder module **608** and symbol distribution module **609** such that the alignment markers appear essentially intact and in order in the two-lane transmit data stream output by symbol distribution module **609**.

[0027] A Reed-Solomon (RS) encoder module **608** operates on 10-bit “symbol” blocks of the data stream **607** from the insertion module **606**, adding redundancy to enable downstream correction of symbol errors. Typically, the encoder module **608** operates to preserve the original data stream content while appending so-called “parity” information, e.g., 30 parity symbol blocks appended to 514 data symbol blocks to form a complete code word block. Thus the alignment blocks inserted by module **606** will remain present in the output data-stream from the encoder module. A symbol distribution module **609** distributes code word symbols across multiple transmission lanes in round-robin fashion, directing each lane to a corresponding transmitter. Optionally, an alignment marker detection module **610A** (and **610B**) for each lane of the transmit data stream detect the alignment markers inserted by module **606** with suitable data buffering. Alternatively, the alignment marker detection modules **610A**, **610B** can be omitted in favor of appropriate direct signaling from the alignment marker insertion module **606**. In either case, the control modules **620A**, **620B** control the multiplexers **612A**, **612B** in each lane, based at least in part on detection signals indicating the presence of the alignment markers.

[0028] Multiplexers **612A**, **612B** forward the encoded data streams to serializer modules **614A**, **614B** during normal operations and in the absence of alignment markers. During auto-negotiation and training phases, the multiplexers supply negotiation and training data streams from the training control modules **620A**, **620B** to the serializers. During normal operations in the presence of alignment markers, the multiplexers **612A**, **612B** (acting as alignment marker replacement modules) may supply the serializer modules with modified alignment markers to convey hidden channel information. The hidden channel can convey management bus data from the MCU in one end connector to the DRR device(s) in the other end connector(s), and optionally may convey adaptation information for the pre-equalizers. The serializers **614A**, **614B**, each accept a stream of transmit data blocks and convert the stream of blocks into a (higher-rate) stream of channel symbols. Where, for example, a 4-PAM signal constellation is used, each serializer may produce a stream of two-bit symbols.

[0029] Each stream of channel symbols is filtered by a pre-equalizer module **616A**, **616B** to produce a transmit signal, which is amplified and supplied to the transmit channel by a driver **618A**, **618B**. The pre-equalizer modules compensate for at least some of the channel dispersion, reducing or eliminating the need for receiver-side equalization. Such pre-equalization may be

advantageous in that it avoids the noise enhancement often associated with receiver-side equalization and enables digital filtering with a reduced bit-width.

[0030] During the training process, the one or more controllers **620A**, **620B**, receive backchannel information extracted by the receiver from the received data stream. The backchannel information may include management bus information to be conveyed to the local MCU, and may further include adaptation information that the controller can use to adjust the coefficients of the pre-equalization filters. The controllers further receive management bus information from the local MCU to be conveyed to the remote MCU(s), and “remote adaptation info”, which includes locally-generated information for adapting the coefficients of the pre-equalization filter in the remote node. Based on this information the controllers populate the relevant fields of the training frames to provide backchannel information to the remote node. As training frames are employed only during the training phase, and as it may be desirable to continue updating the pre-equalization filter during normal operations, the controller(s) **620** may include similar hidden channel information in or with the modified alignment markers supplied via multiplexers **612A**, **612B** during normal operations.

[0031] We now turn to FIG. 7, which shows a receive chain obtaining analog electrical signals from different receive channels (indicated by Lane0-rx and Lane1-rx). These may be obtained directly from electrical conductors, if the physical medium is an electrical bus or cable, or indirectly via transducers if the physical medium is wireless. Optional low noise amplifiers (LNA) **702A,B**, each provide a high input impedance to minimize channel loading and amplifies the receive signal to drive the input of a continuous time linear equalizer (CTLE) filter **704A,B**.

[0032] CTLE filters **704A,B** provide continuous time filtering to shape the receive signal spectrum in an adaptive fashion to reduce the length of the channel impulse response while minimizing leading inter-symbol interference (ISI). Decision feedback equalizers (DFE) **706A,B** operate on the filtered signals to correct for trailing ISI and detect each transmitted channel bit or symbol, thereby producing a demodulated digital data stream. Some embodiments employ oversampling. Clock recovery and adaptation modules **708A,B** derive a sampling clock signal from the input and/or output of the DFE's decision element and supply it back to the DFE to control timing of the symbol detection. The adaptation modules **708A,B** further derive an error signal of the DFE decision element's input relative to the output or (during the training phase) to a known training pattern, and use the error signal to adapt the DFE coefficient(s) and the response of the CTLE filters. The adaptation modules still further use the error signal to generate “remote info”, i.e., adaptation information for the remote pre-equalizers. This remote info is supplied to the controller(s) **620** (FIG. 6).

[0033] Deserializers **709A,B** group the digital receive data stream bits or symbols into blocks to enable the use of lower clock rates for subsequent on-chip operations. Alignment marker detection modules **710A**, **710B** monitor the receive data stream to detect the CM pattern of the alignment markers and achieve alignment marker lock during normal operations, or during training operations to detect the training frame markers and achieve lock thereto. The backchannel information extraction modules **711A**, **711B** extract the backchannel information from the appropriate portions of the training frames and alignment markers, providing the pre-equalizer adaptation information and status report information to the controller(s) **620**. During normal operations, the multiplexers **712A**, **712B**, operating under control of the extraction modules **711A**, **711B**, replace the modified alignment markers with sets of grouped PCS alignment markers, thereby hiding the hidden channel information fields from the higher layers. The detection modules **710** may communicate the alignment marker detection information to the FEC lane deskew module **713**.

[0034] During normal operations, the receive data streams from the deserializers are aligned by an FEC lane deskew module **713**. If the FEC lanes have somehow gotten switched, an FEC lane reordering module **714** detects the switch based on the contents of the alignment markers and compensates for the switch as the two lanes are multiplexed on a symbol-block basis into a single lane. An RS decoder module **716** operates on the single lane data stream to detect and correct any

symbol errors, removing the FEC coding redundancy (parity symbols) during the decoding process. [0035] A transcoding module **720** converts the 256b/257b transmission code words into blocks of four 64b/66b transmission code words distributing the 66-bit code word blocks across four PCS lanes. An alignment marker insertion module **724** converts the sets of grouped alignment markers into individual alignment markers having lane-specific UM patterns, and inserts the individual alignment markers at appropriate positions in the four lanes, accounting for the operation of the transcoding module **720**. The multi-lane PCS data stream is provided to the higher hierarchy layers of the node for eventual communication of the conveyed data to the destination application.

[0036] Multiple such receive and transmit chains may be included within each of the DRR devices. The DRR devices provide buffering between the receive and transmit chains, optionally providing format conversion e.g., between NRZ and PAM4 signal constellations, between symbol rates, and/or between numbers of data lanes. Data stream steering/broadcasting may be provided from the non-redundant end connector to the redundant end connectors, and data stream selection/multiplexing provided from redundant end connectors to the non-redundant end connector.

[0037] Whether the remote end connector accessibility is provided by a serial management bus between local and remote MCUs, by a direct I2C bus from the local connector to the remote MCU, or by a hidden channel between DRR devices, it enables the cable to be monitored and controlled from a single end rather than requiring each end of an N-ended cable to participate. Where only one MCU is present, each host can provide commands to it to retrieve and update register information, thereby determining status, performance, parameter settings, and even update firmware. Where multiple end connectors each have an MCU, each host can address commands to any selected one of the MCUs, enabling each one of them to retrieve and update register information from all the MCUs.

[0038] Various mechanisms are contemplated to cope with any potential latency when the MCU in a local end facilitates communication with the MCU in a remote end. One contemplated mechanism is the real-time read, in which the local MCU receives the read command but delays any response until the requested data has been fetched from the remote MCU, using clock stretching or a similar technique to stall for time.

[0039] Another contemplated mechanism is the delayed read request, in which the local MCU determines that the requested data is not available locally and sends an acknowledgement of the request. The local MCU then fetches the requested data to a local cache and sends a notification to the host that the data is ready to be read. The host can then repeat the request and the local MCU can respond with the requested data from the local cache.

[0040] A third contemplated mechanism is for the local MCU to obtain relevant data from the remote MCUs either periodically or as it gets updated, caching the obtained data for immediate availability for read commands on the management bus.

[0041] The remote end connector accessibility greatly facilitates cable performance monitoring and maintenance in data center environments. Rather than forcing each host port to participate, server and switch alike, the monitoring and maintenance services can optionally be consolidated within the switches, thereby eliminating any need to provide the firmware image to each network node and greatly reducing bandwidth consumption. Alternatively, the services can be distributed to the server nodes to minimize any added processing burden on the switches. As yet another alternative, each host port can provide the service to add redundancy to the process, ensuring each cable is monitored and maintained even if one of its hosts happen to be disconnected or shut down when the process is performed.

[0042] As one example, FIG. **8** is a flow diagram of an illustrative method for upgrading cable firmware in a remote end connector of a cable. Beginning in block **802**, the host checks whether the cable coupled into a local host port is offline, and if so, the host initiates the power-up process for the cable in block **804**. In block **806**, the host sends an inquiry via the local management bus

connector pins to the MCU to obtain information regarding the cable model and version of the installed firmware. If the information indicates that multiple MCUs are present, the host addresses a similar inquiry to each of the MCUs to determine the version of their installed firmware.

[0043] In block **808**, the host compares the firmware version numbers to that known to be the latest version number for that model, or otherwise determines whether the installed firmware version is acceptable. If so, the process completes successfully. Otherwise, in block **810**, the host uses the management bus to write a firmware image to an inactive nonvolatile memory slot of the local MCU. A check is periodically performed in block **812** to determine whether the transfer is complete, and once it is completed, the host requests a checksum from the MCU in block **814** to confirm that the transfer was successful. If not, blocks **812-814** are repeated.

[0044] In block **816**, the host determines whether the cable has any remote MCUs. If not, the host configures the MCU(s) to use the nonvolatile memory slot with the upgraded firmware in block **824**, performs a restart of the cable, and returns to block **802**.

[0045] If one or more remote MCUs is present, in block **818** the host instructs the local MCU to send a copy of the new firmware image to each of the remote MCUs, using the serial bus, the shared management bus, the hidden channel, or some other suitable form of internal channel between the end connectors. The host periodically checks in block **820** to determine whether the internal transfer(s) are complete. Once the transfer(s) have completed, the host requests a checksum from each of the remote MCUs in block **822** to confirm that the transfer(s) were successful. If not successful, blocks **818** to **822** are repeated. Once successful, the host in block **824** configures each of the MCUs to use the upgraded firmware on reboot, and initiates a cable restart. The process resumes in block **802**.

[0046] Numerous alternative forms, equivalents, and modifications will become apparent to those skilled in the art once the above disclosure is fully appreciated. Although only two-ended and three-ended cables were shown in the figures, the disclosed principles are applicable to different cable configurations including those with more than three ends, e.g., a 1:4 breakout cable. The foregoing discussion focused on cables employing DRR devices, as such cables tend to have a greater degree of configurability. Nevertheless, the disclosed principles are also applicable to other active cables having programmable parameters for configurable operation such as those providing analog amplification, analog equalization, and/or analog conversion between electrical and optical signaling. It is intended that the claims be interpreted to embrace all such alternative forms, equivalents, and modifications that are encompassed in the scope of the appended claims.

Claims

1.-16. (canceled)

17. A method that comprises: downloading firmware from a host to a local controller in a local end connector of a cable, the local controller storing the firmware in a local memory; transferring a copy of the firmware via an internal channel from the local controller to a remote controller in a remote end connector of the cable, the remote controller storing the firmware in a remote memory in the remote end connector; and configuring the local controller and the remote controller to each boot from the firmware when the cable is restarted.

18. The method of claim 17, further comprising transferring a copy of the firmware via an internal channel from the local controller to a second remote controller in a second remote end connector of the cable, the second remote controller storing the firmware in a second remote memory.

19. The method of claim 17, further comprising, prior to said downloading, accessing at least one of the local controller and the remote controller to determine whether an installed firmware version is current.

20. The method of claim 17, further comprising initiating a restart of the cable.

21. A network that comprises: multiple network nodes including one or more switches; active

network cables that interconnect the multiple network nodes, each active network cable having a local end connector coupled to a host port of the one or more switches and having a remote end connector, each of the switches being configured to, in each connected one of the active network cables, perform a firmware update by: downloading firmware to a local controller in the local end connector, the local controller storing the firmware in a local memory; initiating copying of the firmware via an internal channel from the local controller to a remote controller in the remote end connector, the remote controller storing the firmware in a remote memory in the remote end connector; and configuring the local controller and the remote controller to each boot from the firmware when restarted.

22. The network of claim 21, wherein no non-switch network nodes are configured to perform the firmware update.

23. The network of claim 21, wherein as part of performing the firmware update each of the switches is further configured to initiate copying of the firmware via the internal channel from the local controller to a second remote controller in a second remote end connector of the cable, the second remote controller storing the firmware in a second remote memory.

24. The network of claim 21, wherein as part of performing the firmware update each of the switches is further configured to access at least one of the local controller and the remote controller to determine whether an installed firmware version is current.

25. The network of claim 21, wherein as part of performing the firmware update each of the switches is further configured to initiate a restart of that cable.

26. A switch that comprises: multiple ports each configured to couple with a local end connector of a cable, the switch providing a maintenance service that performs a firmware update for each active network cable coupled to a respective one of the multiple ports by: downloading firmware to a local controller in the local end connector, the local controller storing the firmware in a local memory; initiating copying of the firmware via an internal channel from the local controller to a remote controller in the remote end connector, the remote controller storing the firmware in a remote memory in the remote end connector; and configuring the local controller and the remote controller to each boot from the firmware when restarted.

27. The switch of claim 26, wherein as part of performing the firmware update the maintenance service initiates copying of the firmware via the internal channel from the local controller to a second remote controller in a second remote end connector of the cable, the second remote controller storing the firmware in a second remote memory.

28. The switch of claim 26, wherein as part of performing the firmware update the maintenance service accesses at least one of the local controller and the remote controller to determine whether an installed firmware version is current.

29. The switch of claim 26, wherein as part of performing the firmware update the maintenance service initiates a restart of that cable.
